

AI Smart Document Assistant - Project README

Project Objective

Build an AI assistant using FastAPI, Groq LLM, LangChain, LangGraph, ChromaDB and RAG. Users can upload PDFs, ask questions, use calculator tool and memory.

Technologies

Python, FastAPI, Groq API, LangChain, LangGraph, ChromaDB, Sentence Transformers, PyPDF, Pydantic.

Features

Chat API, Upload PDF, RAG, Memory, Calculator Tool Calling, AI Agent, LangGraph.

Architecture

User -> FastAPI -> AI Agent -> Chat / RAG / Calculator -> Response

RAG Flow

PDF -> Chunks -> Embeddings -> ChromaDB -> Retrieval -> Groq LLM -> Answer

Endpoints

GET /, POST /chat, POST /upload, POST /ask, POST /remember, GET /recall/{key}, POST /calculate, POST /langgraph

Concepts

LLM, RAG, Embeddings, Vector Database, Tool Calling, Memory, AI Agent, LangGraph.

Future

Multiple PDFs, React Frontend, Persistent Memory, Deployment.

Developer

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